Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/12/2010
PI Name:	Yang, Xianbin Ph.D.		
Project Title:	Thioaptamer Diagnostic System		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	(832) 379-2175
Organization Name:	AM Biotechnologies, LLC		
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PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77034-4509	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	06/28/2010	**End Date:**	12/27/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10CA98C		
Performance Goal No.:			
Performance Goal Text:			
	AM Biotechnologies (AM) in partnership with Sandia National Laboratories will develop a Thioaptamer Diagnostic System (TDS) in response to Topic X10.01 Reusable Diagnostic Lab Technology. The TDS will quickly quantify clinically relevant biomarkers in flight using only microliters of virtually any complex sample. The system combines ambient-stable, long-shelf-life affinity agent assays with a handheld microfluidic gel electrophoresis affinity assay quantification technology. The system is easy to use, compatible with operation in microgravity, and designed to permit simultaneous quantification of 32 or more biomarkers from a single astronaut sample. Phase 1 of this project demonstrated that a thioaptamer assay used in the microfluidic instrument can quantify a specific biomarker in serum in the low nanomolar range. AM also successfully identified novel affinity agents to bone specific alkaline phosphatase (BAP) and demonstrated their use to detect BAP using the microfluidic instrument. Phase 2 will expand the number of ambient stable affinity agents and demonstrate a TDS prototype to NASA. AM anticipates that		

Task Description:	the TDS at the end of Phase 2 will be at TRL 4 to 5. In Phase 3, AM and Sandia will produce flight units for NASA research use on the International Space Station (ISS) as well as for diagnostic use on future long duration missions. POTENTIAL NASA COMMERCIAL APPLICATIONS: Successful development of this system will provide NASA with both near-term as well as long-term advantages. Near-term, a non-clinical version of the system will be extraordinarily useful for conducting research on ISS into human adaptation to microgravity. The TDS could monitor levels of numerous biomarkers in astronauts on orbit and would eliminate the need to freeze samples on orbit and return them to Earth for ground-based testing. Long-term, the clinical version of the TDS will provide a robust, flight tested diagnostic capability for missions to the moon and Mars. Virtually any type of ambient stable assay that is of interest to NASA could be developed for the system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Thioaptamer Diagnostic System will be commercialized first in the research support market, and then later as an FDA-approved hand-held clinical diagnostic system for non-traditional and resource poor locations. Clinical TDS units will be targeted for use in physician offices and by emergency responders. Research support market units will be targeted at lower-throughput biomedical research laboratories in universities and companies with limited capital budgets that have limited or no access to core support laboratories.
Task Progress:	New project for FY2010. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)